

Practice: Classify the System

For each matrix, you must decide:

- Consistent or Inconsistent
- Unique / Infinite / No solution

Q1

$$\begin{bmatrix} 1 & 1 & | & 2 \\ 0 & 1 & | & 1 \end{bmatrix}$$

Q2

$$\begin{bmatrix} 1 & 2 & | & 4 \\ 2 & 4 & | & 8 \end{bmatrix}$$

Q3

$$\begin{bmatrix} 1 & 1 & | & 2 \\ 1 & 1 & | & 5 \end{bmatrix}$$

Q4

$$\begin{bmatrix} 1 & 0 & 2 & | & 3 \\ 0 & 1 & 1 & | & 2 \\ 0 & 0 & 1 & | & 1 \end{bmatrix}$$

Q5

$$\begin{bmatrix} 1 & 2 & 3 & | & 6 \\ 2 & 4 & 6 & | & 12 \\ 1 & 1 & 1 & | & 3 \end{bmatrix}$$

Q6

$$\begin{bmatrix} 1 & 2 & 3 & | & 6 \\ 2 & 4 & 6 & | & 10 \\ 1 & 1 & 1 & | & 3 \end{bmatrix}$$

Q7

$$\begin{bmatrix} 1 & 0 & 1 & | & 2 \\ 0 & 1 & -1 & | & 1 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

Q8

$$\begin{bmatrix} 1 & 0 & 1 & | & 2 \\ 0 & 1 & -1 & | & 1 \\ 0 & 0 & 0 & | & 5 \end{bmatrix}$$

Q9

$$\begin{bmatrix} 1 & 2 & 0 & | & 3 \\ 0 & 1 & 4 & | & 2 \\ 0 & 0 & 0 & | & 0 \end{bmatrix}$$

What you are asked to.....

For each:

1. Count **pivots** → **rank**
2. Count **variables**
3. Check last row
4. Decide:
 - Consistent / Inconsistent
 - Unique / Infinite / No solution